

Solution

LIGHT -09

Class 10 - Science

Section A

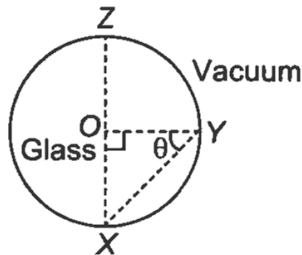
1.

(d) Red

Explanation: Violet has the shortest wavelength and red has the longest wavelength.

2. (a) $2.4 \times 10^8 \text{ m s}^{-1}$

Explanation: This is a case of total internal reflection.



Here $\theta = 45^\circ$ and $\theta > \theta_c$

$$\Rightarrow 45^\circ > \theta_c \Rightarrow \sin 45^\circ > \sin \theta_c$$

$$\Rightarrow \frac{1}{\sqrt{2}} > \frac{1}{\mu} \Rightarrow \mu > \sqrt{2} \quad \left(\text{As } \sin \theta_c = \frac{1}{\mu} \right)$$

$$\Rightarrow \frac{c}{v} > \sqrt{2}$$

$$v < \frac{c}{\sqrt{2}} = \frac{3 \times 10^8}{\sqrt{2}} \Rightarrow v < 2.1 \times 10^8$$

Only $2.4 \times 10^8 \text{ m s}^{-1}$ is not possible.

3.

(c) Beam of light

Explanation: A bundle of light rays is called a beam of light. When source of light emits light, parallel beam of light together forms beam of light.

4.

(c) is directly proportional to the thickness of the glass slab.

Explanation: The lateral displacement of an incident ray passing out of a rectangular glass slab is directly proportional to the thickness of glass slab, angle of incidence, and refractive index however it is inversely proportional to the wavelength of the incident light.

5.

(c) lens and the screen

Explanation: The focal length of the lens, image distance should be known which is the distance between the lens and the screen.

6.

(c) Z

Explanation: The lesser the optical density, the more is the speed of light.

7.

(b) always

Explanation: always

8.

(c)

$$r = \sin^{-1} \left(\frac{1}{n} \right)$$

Explanation: According to Snell's law of refraction, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant for a given pair of media, and for the light of a given colour. If "i" is the angle of incidence and "r" is the angle of refraction, then:

$$\frac{\sin i}{\sin r} = n$$

$$\Rightarrow r = \sin^{-1}\left(\frac{1}{n}\right)$$

9.

(c) light from sun

Explanation: Larger the distance more sharper the image.

10. **(a)** Clay

Explanation: Clay cannot be used to make a lens. Only a transparent material can be used to make a lens.

11.

(b) The difference in the wavelength

Explanation: The deviation in the path of the light is inversely proportional to the wavelength. Because different wavelengths (colors) of light travel through a medium at different speeds, the amount of bending is different for different wavelengths. Violet is bent the most and red the least because violet light has a shorter wavelength, and short wavelengths travel more slowly through a medium than longer ones do. The dispersion of light takes place due to differences in the wavelength of different colours of light.

12.

(d) carbon disulphide

Explanation: carbon disulphide

13.

(c) $e = i$

Explanation: For parallel surfaces, $e = i$ should be obeyed.

14.

(c) full image of reduced brightness is formed

Explanation: There will be a drop in the intensity due to less light crossing the lens, but full image is formed as the light from every point of the object will reach the lens.

15.

(b) virtual, behind the mirror, and of the same size as the object

Explanation: virtual, behind the mirror, and of the same size as the object

16.

(b) $180 - 2\theta$

Explanation: Since $\angle r = \angle i$, the deviation should be $(180 - 2\theta)$.

17.

(b) increases

Explanation: The lateral displacement will increase if a glass block is made thicker as lateral displacement is directly proportional to the thickness of the glass slab.

18.

(b) $\frac{f_1 \times f_2}{f_1 + f_2}$

Explanation: The net power of the lenses placed in contact is given by the algebraic powers of the individual powers. Equivalent power of the lens combination $P = P_1 + P_2$

$\therefore \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$ where f_1 and f_2 are the focal lengths of the two lenses, and F is the focal length of the combination.

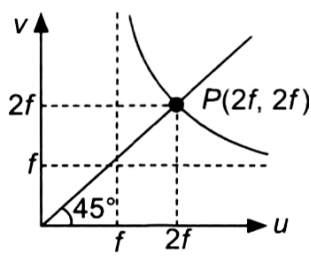
$$\frac{1}{F} = \frac{f_1 + f_2}{f_1 \times f_2}$$

$$\Rightarrow F = \frac{f_1 \times f_2}{f_1 + f_2}$$

19.

(d) $(2f, 2f)$

Explanation: Consider the case of a convex lens.



For a lens, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

So, $v = \infty \Rightarrow |u| = f$ and $u = \infty \Rightarrow |v| = f$

Hence, the curve will be as shown.

Also, for the given line, at every point $v = u$.

So, the line and curve can meet at $u = 2f$ position only.

So, the coordinates of P will be $(2f, 2f)$.

20.

(d) $\frac{1}{3}$

Explanation: $\frac{1}{3}$

21. (a) Either plane or convex

Explanation:

The mirror may be a plane mirror or a convex mirror. A plane mirror always forms an erect image. A convex mirror always forms an erect, and a virtual and diminished image, for any position of the object.

22.

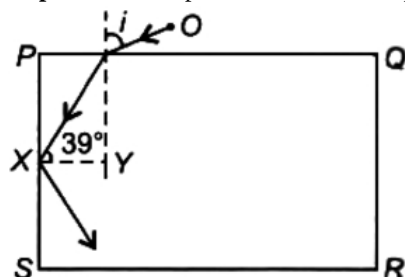
(b) in medium C

Explanation: in medium C

23.

(b) None of these

Explanation: A pin is held at O, mid point of PQ.



When angle of incidence becomes greater than critical angle, the ray starting from O will undergo total internal reflection and pin shall not be seen at all.

24.

(c) All of these

Explanation: The focal length of the convex lens depends on the refractive index and radius of the surface (curvature) mainly. The color of the light passing through the lens is not a major factor but still, it counts.

25.

(c) converges at focus (f)

Explanation: The point of convergence of parallel beams from a distant object is called the focus.

Section B

26. (a) neither the feet nor the heads of the images of pins P_3 and P_4 are in line with the feet and heads, respectively, of pins P_1 and P_2 .

Explanation: The emergent ray drawn by him is not parallel to the incident ray

27.

(b) A and the device X is a convex lens

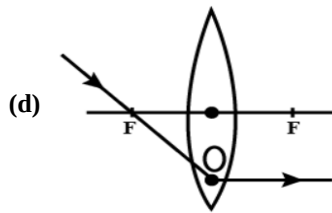
Explanation: Inverted, sharp and real image of distant tree is formed by a convex lens.

28.

(b) MS

Explanation: MS

29.



Explanation: In the convex lens, incident ray passing through F goes parallel to the principal axis after refraction.

30. (a) A rectangular glass slab

Explanation:

A rectangular glass slab could be inside the box since the emergent rays are parallel to the incident rays. A convex lens converges the parallel rays while a concave lens diverges the parallel rays. Parallel rays incident on the face of a prism will bend towards its base.

31.

(b) 11.0 cm

Explanation: Distance of pole to focus is called focal length :

$$\therefore f = 15.6 - 4.6 = 11.0 \text{ cm}$$

32.

(d) C

Explanation: The best result will be obtained by the student C because of the following reasons:

- i. The distant object is clearly visible.
- ii. The image formed on the screen is well defined and sharp.
- iii. Student C kept the measuring scale parallel to the ground that helps in accurate measuring of the focal length (distance) of a concave mirror.

In the case of students A, B, and D, a blurred image is formed which gives the incorrect measurement of the focal length (distance) of a concave mirror.

33.

(b) IV

Explanation: All parallel beams are to pass through the focus.

34.

(b) Concave mirror

Explanation: Because the screen is on the same side of the object which means it is never a lens becoz it happens behind the lenses in such case. Moreover Concave mirror forms real images i.e. image can be obtained on a screen.

35.

(c) Convex lens

Explanation: Since a convex lens can converge the light rays at a point and emerged at point B. So, the convex lens is inside the box.

36.

(b) NC

Explanation: For parallel surfaces the emergent and incident rays should be parallel.

37.

(a) B

Explanation: Snell's law gives the relationship between angles of incidence and refraction for a wave impinging on an interface between two media with different indexes of refraction.

Thus, light bends towards normal when it passes from air to glass. Light bends away from normal when it passes from glass to air.

38.

(b) 9.25

Explanation: 9.25

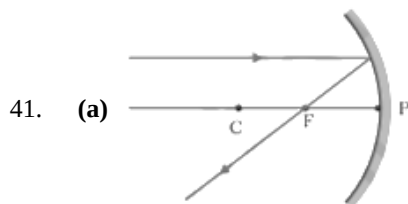
39. (a) 'I' keeps on increasing with increase in angle of incidence.

Explanation: $I \propto \sin(i - r)$ where r is the angle of refraction.

40.

- (b) greater than unity

Explanation: Here the ray of light bends towards normal when it enters from medium A into medium B. This shows that medium A is optically rarer than medium B. Therefore, the speed of light in medium A is more than the speed of light in medium B. So, ratio of speed of light in medium B to speed of light in medium A will be less than one.



Explanation: In the case of a concave mirror, an incident ray is parallel to the principal axis passes through F after reflection.

42.

- (c) X and Z respectively

Explanation: Angle of zero at normal will directly give the measure of the angles of incidence and reflection.

43.

- (d) 2, 1, 4, 3

Explanation: 2, 1, 4, 3

44.

- (d) $\angle i > \angle r$

Explanation: When light is entering from optically rarer to optically denser medium, the angle of incidence will be greater than the angle of refraction.

So, for parallel surfaces, refracting light, $\angle e = \angle i$ and for a denser medium $\angle r < \angle i$.

45.

- (b) S

Explanation: S

46.

- (b) < 1

Explanation: As light enters a rarer medium from a denser medium, it will bend away from the normal.

47.

- (b) D

Explanation: On regular reflection $\angle r = \angle i$.

48.

- (c) $\frac{\sqrt{3}}{\sqrt{2}}$

Explanation: According to Snell's law of refraction, the ratio of the sine of angle of incidence to the sine of angle of refraction is a constant for a given pair of media, and for the light of a given colour. If "i" is the angle of incidence and "r" is the angle of refraction, then:

$\frac{\sin i}{\sin r} = \text{constant}$, where the constant = Refractive medium of the second medium with respect to the first medium.

Given, the angle of incidence = 60° , and the angle of refraction = 45°

Therefore, refractive index of medium B with respect to medium A, $n_{BA} = \frac{\sin i}{\sin r} = \frac{\sin 60^\circ}{\sin 45^\circ} = \frac{\sqrt{3}}{\sqrt{2}}$

49.

- (c) $\theta_1 < \theta_2$ but $\theta_1 = \theta_3$

Explanation: For $i = (90 - \theta_1)$, the value of $r = (90 - \theta_2)$ will be less. So $\theta_2 > \theta_1$.

As long as the surfaces of the slab are parallel the $\angle e = \angle i$

i.e. $(90 - \theta_3) = (90 - \theta_1)$ or $\theta_3 = \theta_1$

Section C

50.

(d) A convex lens has 4 dioptre power having a focal length 0.25 m

Explanation: The power P of a lens of the focal length is given by $P = \frac{1}{f}$, where f is the focal length in meter and power in dioptre

$$P = \frac{1}{f} \text{ or } f = \frac{1}{P} = \frac{1}{4} = 0.25\text{m}$$

The positive value for focal length indicates a convex lens.

51.

(b) (A)

Explanation: Concave mirrors are used by the dentist to see the large images of teeth of patients since concave mirrors form enlarged images. Concave mirrors are also used for shaving purpose. Convex mirrors form diminished images. They are used as rear-view mirrors in cars, motorcycles, scooters, etc since they give a wide field of view.

52.

(d) All of these

Explanation: All the given statements are correct.

- The speed of all colours is the same in air or vacuum i.e. $3 \times 10^8 \text{ ms}^{-1}$. But the speed of all colours is different in a denser medium.
- The refractive index has no unit since it is a ratio. The absolute refractive index of a medium (n_m) is given by $n_m = \frac{c}{v}$ where c is the speed of light in air, and v is the speed of light in the medium.

53.

(b) B and C

Explanation: A ray of light passing from an optically denser medium to an optically rarer medium bends away from the normal, and a ray of light passing from an optically rarer medium to an optically denser medium bends toward the normal.

Section D

54.

(b) 15

Explanation: From observation third, distance of object from lens = distance of image from lens So, radius of curvature, R = 30 cm

Thus, focal length, $f = R/2 = 30/2 = 15 \text{ cm}$

55.

(b) C

Explanation: C has got $\angle e = \angle i$ and $\angle r < \angle i$ as these are to be satisfied for refraction in a glass slab.

56.

(a) Case II

Explanation: A concave mirror is defined as a mirror which is bent inward in the middle. When the face is placed between the concave mirror and a focus, it reflects a magnified image. For the face, concave mirrors reflect even the pores of the skin.

57.

(b) I

Explanation: It is most accurate since $\angle e$ is (nearly) equal to $\angle i$. The correct value of $\angle e = 60^\circ$.

Section E

58.

(c) the screen in the direction of the lens or the lens in the direction of the screen

Explanation: The candle is closer to the lens when compared to the distant lamp. Hence, the image of the candle will be farther away from the lens when compared to the image of the distant lamp. As the image will be more close to focus as the distance of the object is larger.

59.

(d) B

Explanation: The larger separation between the pins will give a better reading for good observation, the angles should be in the ranges $30^\circ - 60^\circ$.

60. (a) C

Explanation: Angles between 30° and 60° are used for clear vision and the pins are fixed at larger separation as possible.

61.

(c) (C)

Explanation: The angle of emergence and angle of incidence will be equal. Emergent ray is parallel to the incident ray along with original direction but it will be laterally displaced to the left of the incident ray.

62.

(d) B

Explanation: The mirror and the screen should be firmly placed for an accurate measure of the separation.

63. (a) 4 m/s

Explanation: In case of plane mirror, the distance is always doubled therefore the speed is in case between the image and the object. So, when driver is reversing his car at a speed of 2 m/s, then the image is also coming closer to the mirror by 2m/s. Thus, it seems that the image of the truck appears to approach the driver at 4 m/s.

64.

(b) slightly nearer to the lens

Explanation: Image will be exactly at focus only when the object is at large distance.

65. (a) an inverted image of the tree at the focus of the lens

Explanation: The screen is just a device or an instrument where the image formed is to be observed. If the screen is removed, the lens will continue to form the image on its focus. Thus, on removing the screen and on looking through the lens in the direction of the object the image of the distant tree is inverted and formed at the focus.

66.

(d) [$\angle r = 20^\circ$, $\angle e = 30^\circ$] and [$\angle r = 25^\circ$, $\angle e = 45^\circ$]

Explanation: $\angle r < \angle i$ and $\angle e = \angle i$ should be satisfied as the light from the air passes through a glass slab.

67. (a) (20 cm, 20 cm) and (inverted, inverted)

Explanation: Since the object is distant, the nature of the image on the screen in case of both the convex lens and the concave mirror will be real, inverted, highly diminished, and point sized.

If the object is distant, then we get its image at focus on the other side of the convex lens. Therefore, d_1 will be 20cm.

In the case of the concave mirror, if the object is distant, then we get its image at the focus in front of the concave mirror. Therefore, d_2 will be 20 cm.

68.

(d) B and D

Explanation: On each of the sides, the normals are not required. Only on one side in each case, he should draw normals. The pins are to be placed little far from each other and not 1 cm.

69.

(c) lens slightly towards the screen

Explanation: lens slightly towards the screen

70.

(c) on the same wall as the window.

Explanation: The experiment may be performed on the same wall as the window to obtain a sharp image since a concave mirror mostly forms images on the same side as that of the object. The object and the image are on the same side of the concave mirror, except when the object is placed between the pole and the focus.

71. (a) towards the screen

Explanation: When you increase the object distance, the image distance decreases, so the lens should be moved towards the screen.

72.

(b) B, C, D

Explanation: Source of light is required. Needle cannot be used with the rest of the three.

73.

(d) A

Explanation: We need to keep the eye far from the glass slab to have a good and proper alignment of the pins.

74.

(c) towards the screen

Explanation: When the image distance increases, object distance decreases. Thus, the distance between the mirror and screen will decrease. So, the mirror should be moved towards the screen.

75. (a) Concave, plane and convex

Explanation: When the object is between F and P of the concave mirror enlarged image is formed behind the mirror. Hence child can see her head bigger in a concave mirror. She can see her body size of the same size because the plane mirror gives an image of the original size. The convex mirror gives diminished images and babies' legs appear smaller.